

TDT4195: Visual Computing Fundamentals

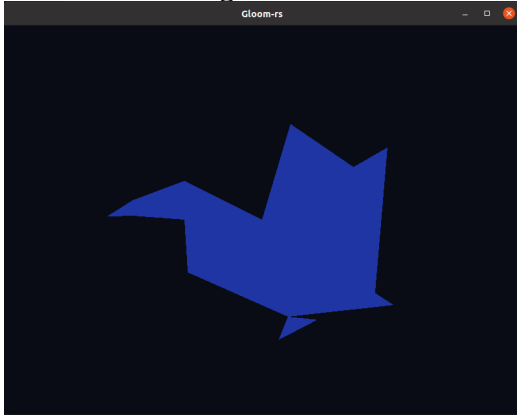
Computer Graphics - Assignment 1

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Task 1c

A VAO containing at least 5 distinct triangles.

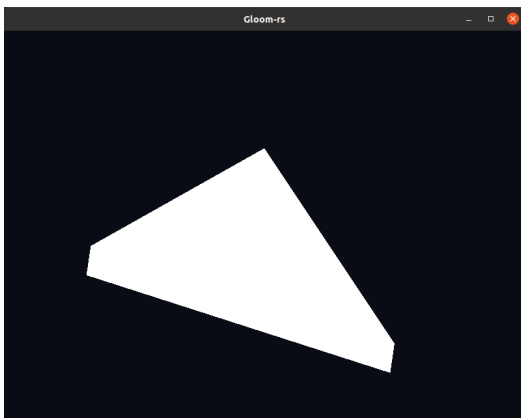


Task 2a

The shape drawn following these vertices:

$$v_0 = \begin{bmatrix} 0.6 \\ -0.8 \\ -1.2 \end{bmatrix}, v_1 = \begin{bmatrix} 0 \\ 0.4 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} -0.8 \\ -0.2 \\ 1.2 \end{bmatrix}$$

will look like this:



The shape does not look like a triangle thanks to the phenomenon called clipping. It occurs when the values in the vertices exceeds the cube around the origin, meaning the values needs to be between '-1' and '1'. Given that two of the points ('-1.2' and 1.2) exceeds this the corner gets cut of when drawn. The Purpose of this function is to ensure objects behind the camera are not visible in the rendered image.

Task 2b

When changing the order in which the vertices of a single triangle is drawn by modifying the index buffer, the triangle either doesn't change or it disappears. The reason it does this is that the viewers

viewpoint is by default supposed to be drawn in a counterclockwise direction. So as an example; the triangle in the code is drawn with [0, 1, 2]. Meaning both [1, 2, 0] and [2, 0, 1] is valid as they follow the same order, but [0, 2, 1], [1, 0, 2] and [2, 1, 0] does not, as they would draw the points in the opposite rotation.

Task2c

the depth buffer is reset for each frame because otherwise all new pixels would be against the depth values from the previous frame. An easy example would be if another object comes into frame and blocks it, then there's no reason to render the previous object.

Assuming we don't use multisampling, the depth test could also make it so that fragments that aren't being rendered could still be processed. A fragment is not the same as a pixel in that multiple fragments can be combined to make one.

The two most common types of shaders are: the vertex shader and the fragment shader. The vertex shader is responsible for: transforming individual vertices around the scene by using; translating, scaling, rotating, etc. It's also responsible for projecting the scene on to the camera. The fragment shader on the other hand is responsible for the color of the fragment. It can also as an added detail render significantly more fragments than there are pixels on the screen.

The index buffer is necessary because when drawing vertices together to form a shape the same vertices can be drawn several times for different triangles. So by combining these together by referring to their index in the VBO, the vertices only have to be drawn once, potentially saving on the memory usage.

If you have multiple entry types in the array buffer, the parameter for the pointer can't be 0.

Task2d

The shape drawn in the code can be flipped by assigning a negative operator to the position. The position is used in the vertices shader.

The color is set up to shuffle between all possible RGB values using time values. The color is defined in the main function, but by using uniform, is sent to the fragment shader.

Task3

The drawn shape is set up to change color constantly over time.